

Small Molecule Structure- Based Drug Design

A crash course to design your own drug computationally

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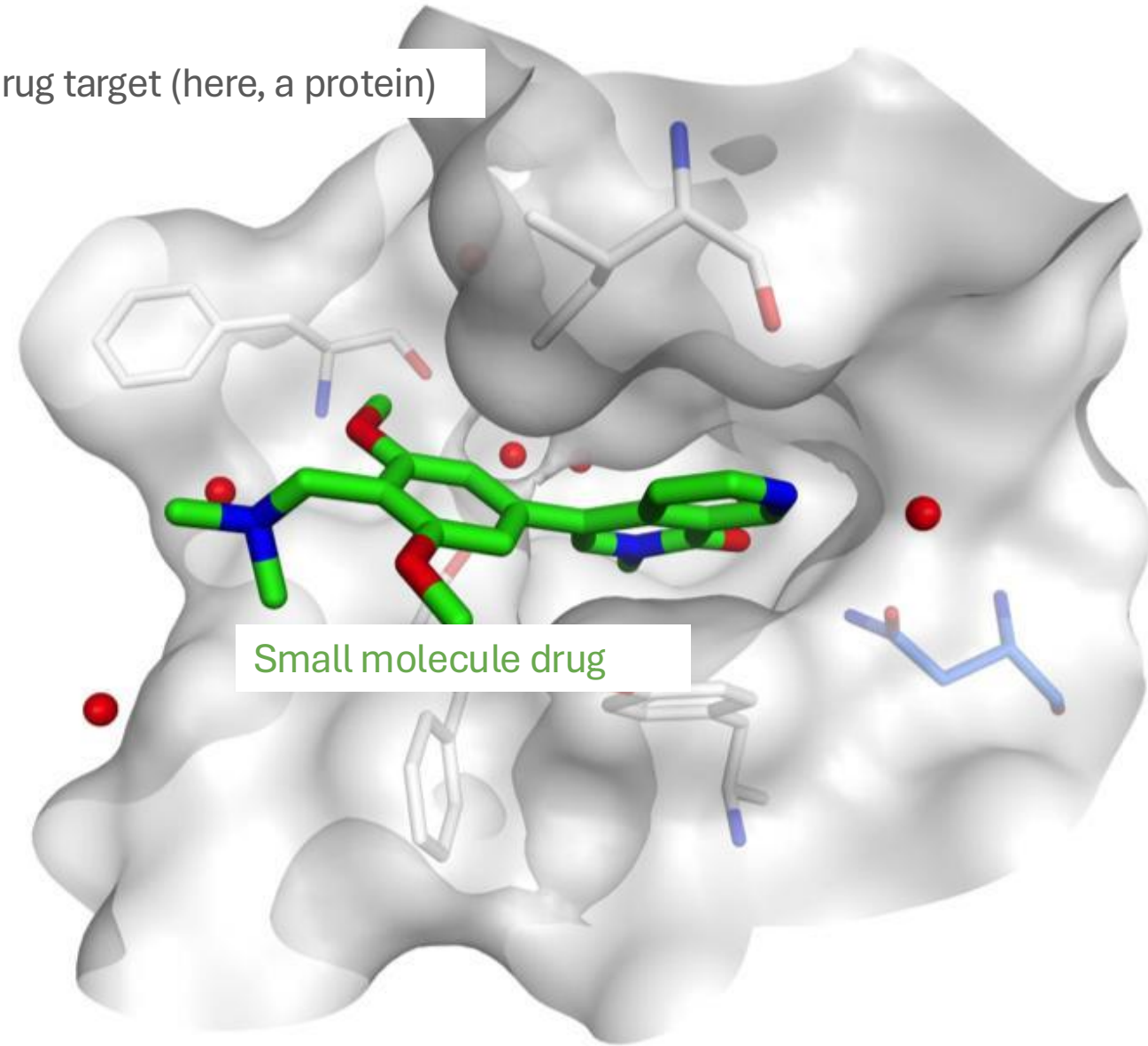
Thursday 13 November 2025

A Refresher on Drug Discovery

What is a drug?

- A **'drug'** is any molecule that **interacts** with a drug target to obtain a **therapeutic effect**
- In pharmaceutical research, a **'drug target'** is a key molecule **implicated** in an infection or **disease**.

Drug target (here, a protein)

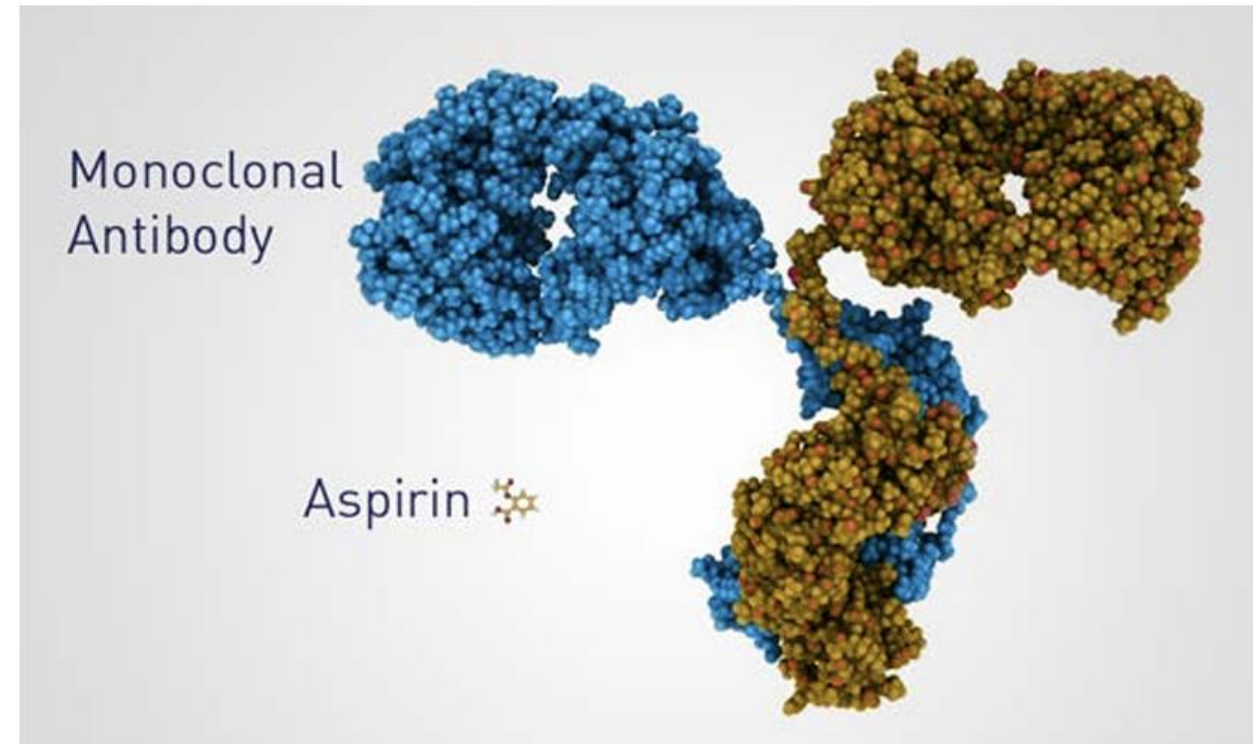


Types of Drugs

Two key classes of drugs:

1. **'Small molecule drugs'** such as aspirin, which are typically produced by chemical synthesis
2. **'Biopharmaceuticals'** or **'biologics'** which are extracted or synthesised from biological sources, such as vaccines

Currently focusing on **small molecule drugs!**



The monoclonal antibody (right) is a large molecule. A single monoclonal antibody weighs more than 800 times what an aspirin molecule (left) weighs.

How were drugs previously discovered?

- Extracts from natural products (usually plants)
- **Empirical/phenotype driven discovery**



Poppy
ca. 1500 BC
“pain relief; sedation”
Opiates



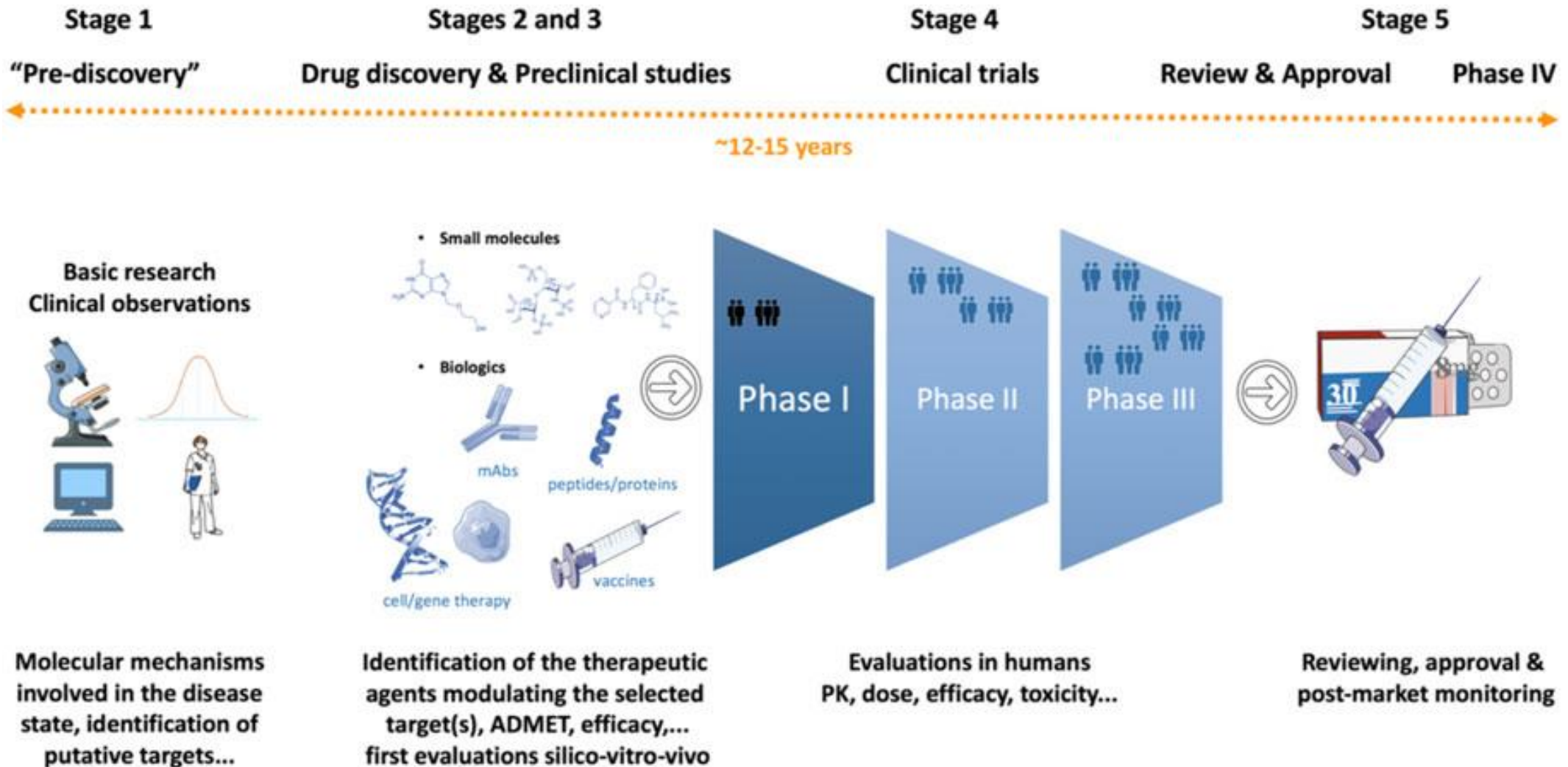
Willow tree extract
ca. 3000-1500 BC
“pain reliever”
Aspirin



Artemisia annua L.
ca. 168 BC
“fever”
Artemisinin

How are drugs discovered today?

(Mostly) target driven discovery

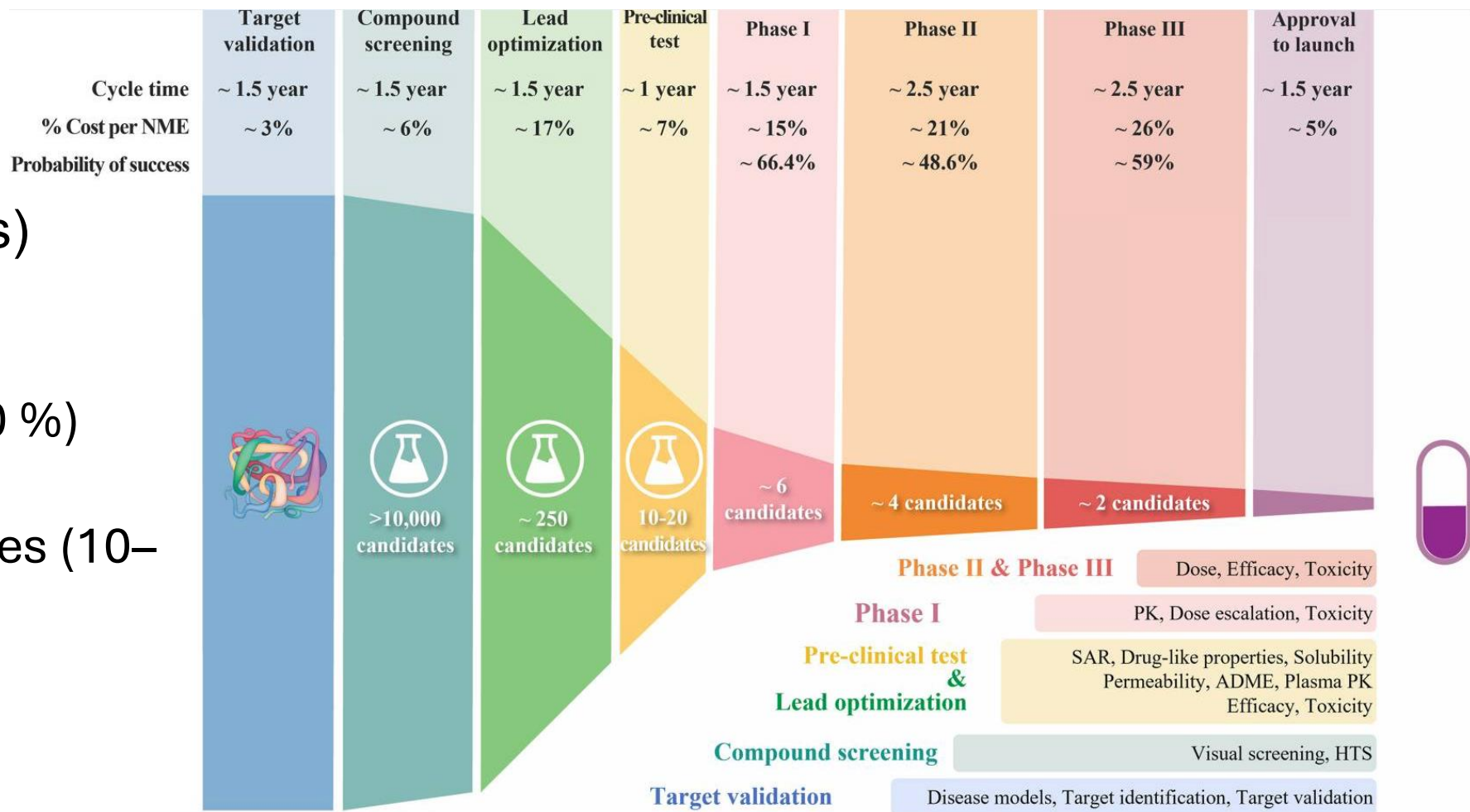


Drug discovery is a risky business

90% failure after Phase I
(beginning of clinical trials)

Main causes:

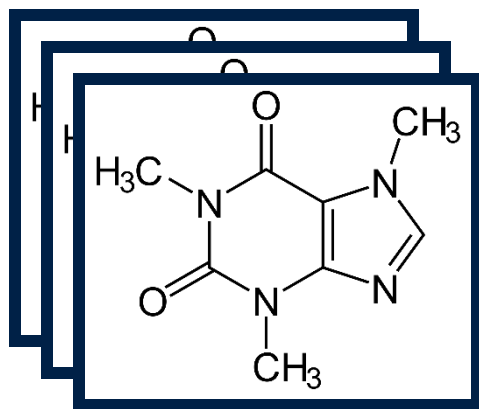
- 1 Lack of efficacy (40–50 %)
- 2 Toxicity (30 %)
- 3 Poor drug-like properties (10–15%)
- 4 Commercial/business factors (10 %)



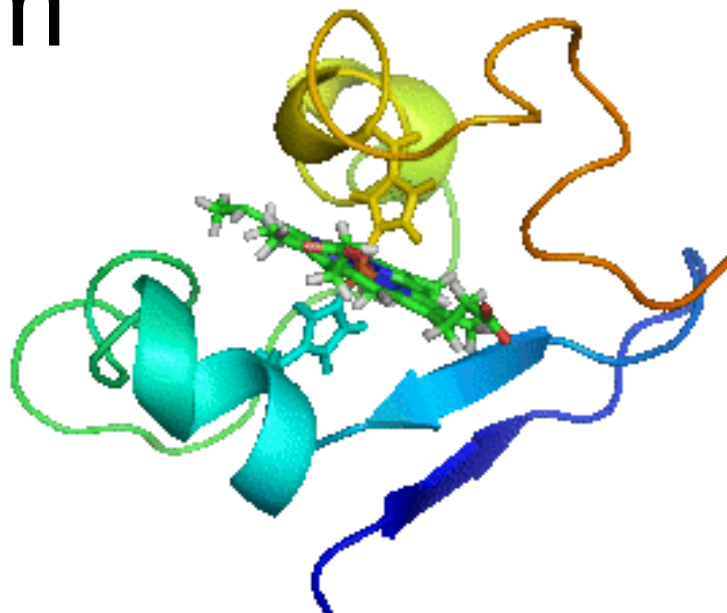
We can use computational methods to maximize success

	Problem	How computational methods can help
1	The drug results in no therapeutic effect (lack of efficacy)	Predict efficacy/affinity/etc.
2	Too toxic!	Predict off-target effects
3	Poor drug-like properties	Predict properties
4	Commercial issues	Business... not relevant

Computers let us test molecules before we make them



Billions of
molecules



Predict properties



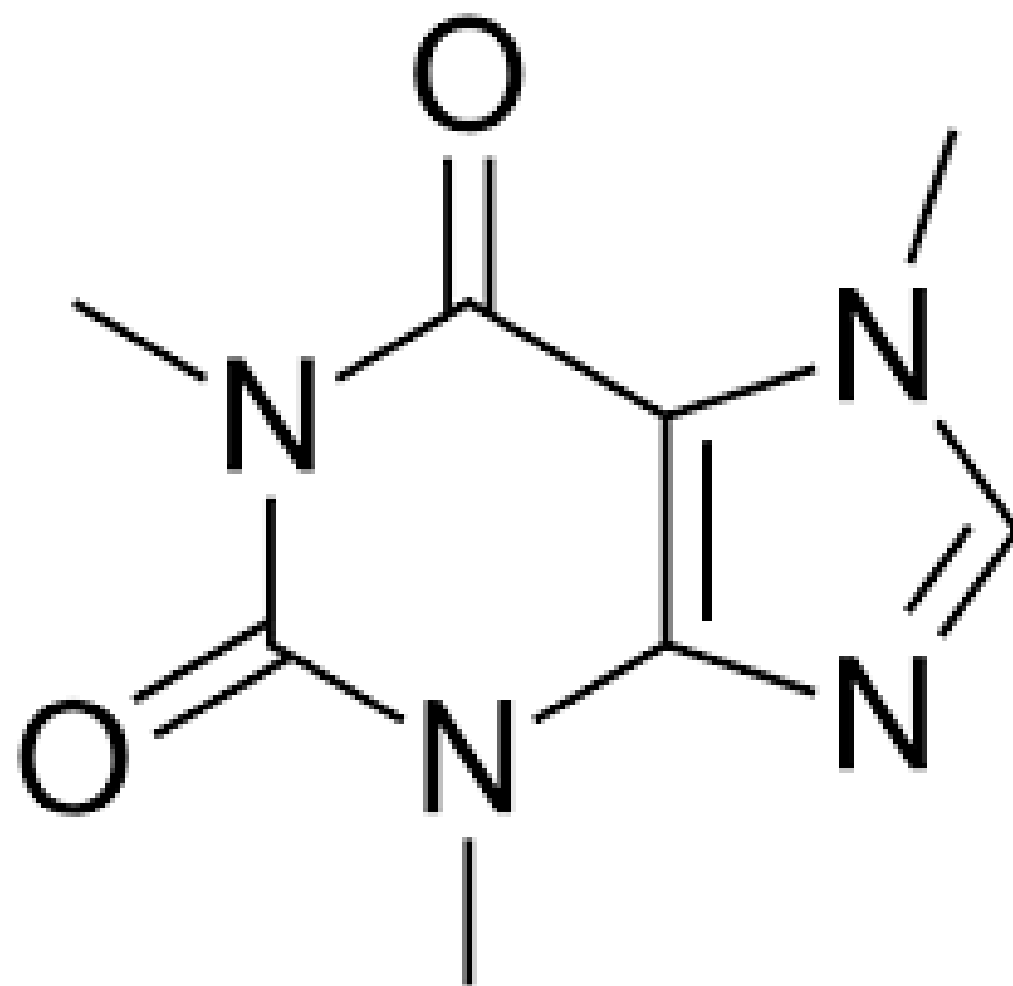
Make best ideas

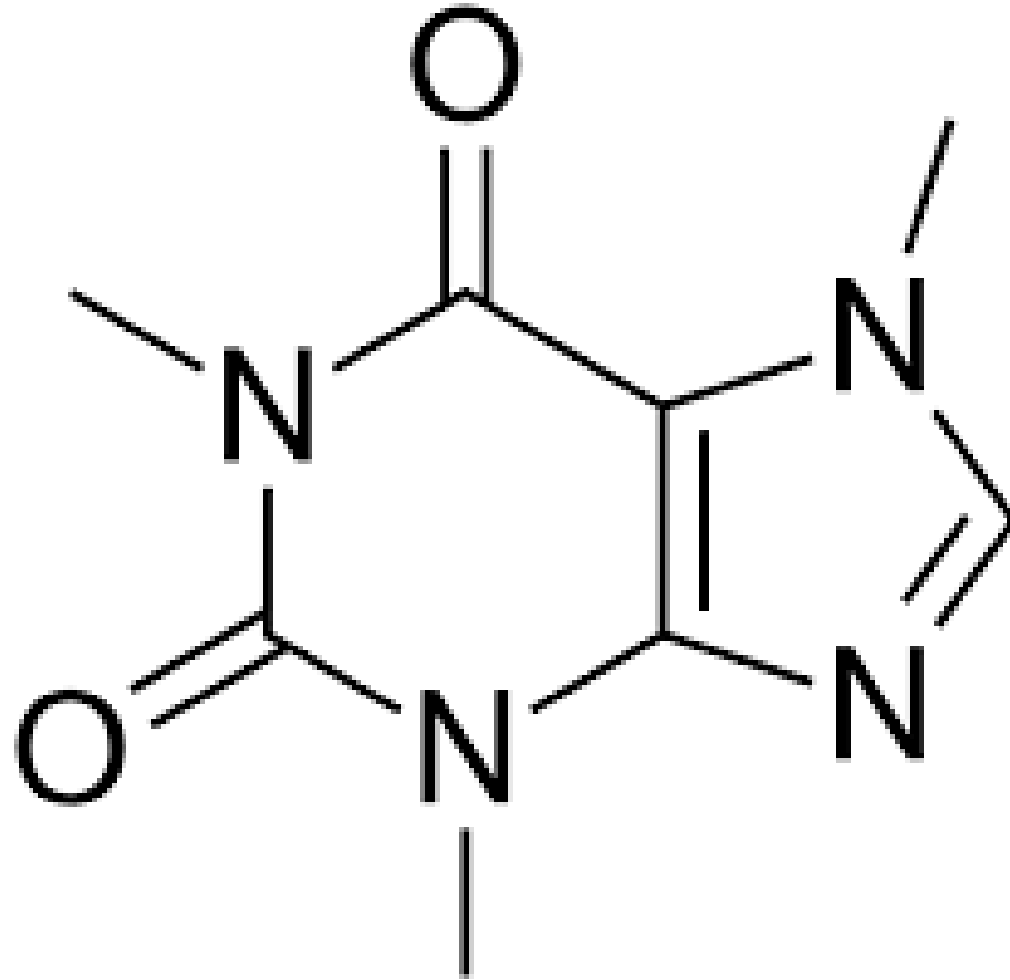
Basic Cheminformatics

How do we computationally represent molecules?

How can we use that representation to predict the properties we want?

1D and 2D Representations



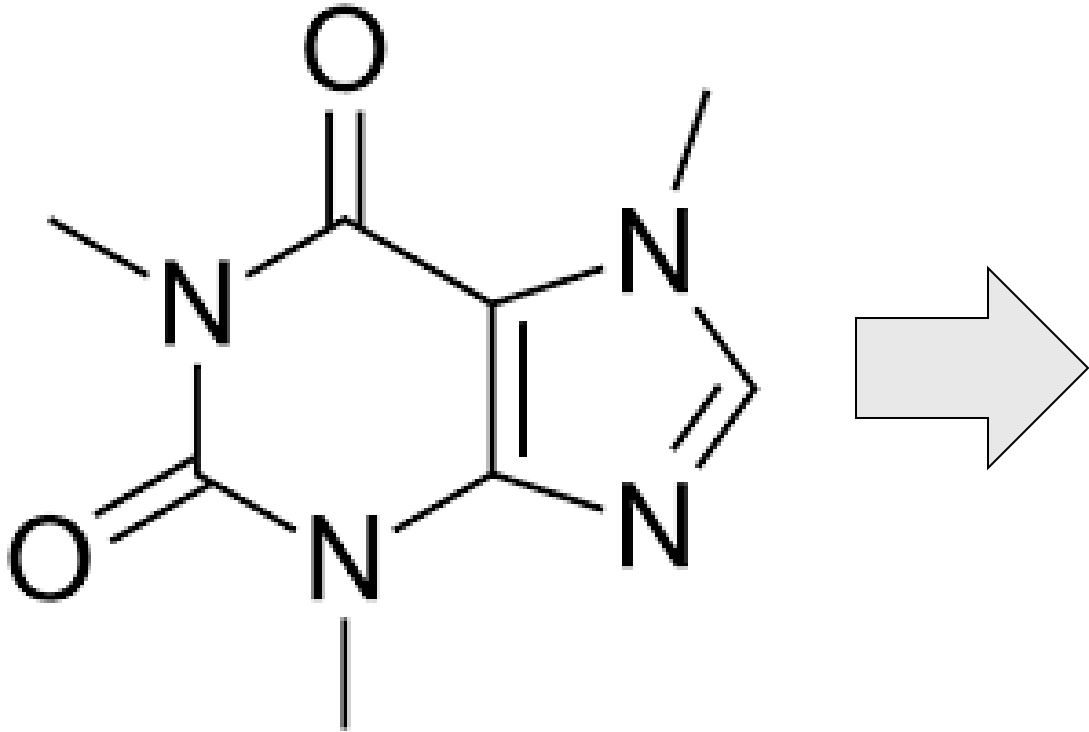


Caffeine

Molecules need to be represented for a computer

1D Representations

Common Name: Caffeine



IUPAC: 1,3,7-Trimethylpurine-2,6-dione

SMILES:

CN1C=NC2=C1C(=O)N(C(=O)N2C)C

Fingerprint (MACCS):

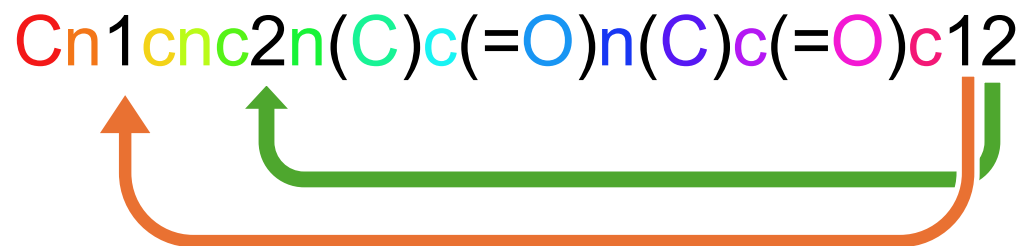
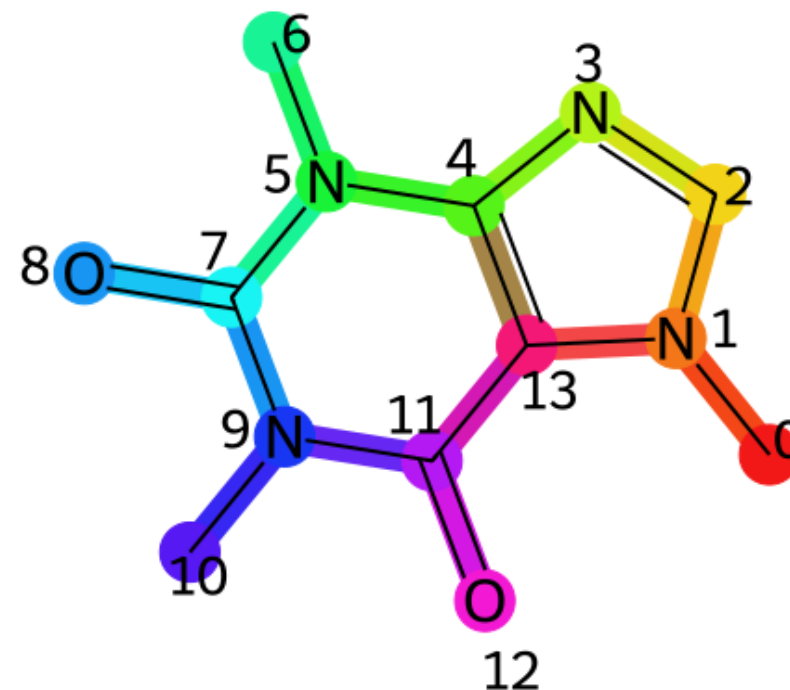
```
00000000000000000000000000000000001100000000000000000000000000000010000000001  
01011001010001001101111001000110001001000100111001010000000011000111100011100  
010101111111110
```

IUPAC = International Union of Pure and Applied Chemistry

SMILES strings are a text representation of the molecular graph

Intuitive characters

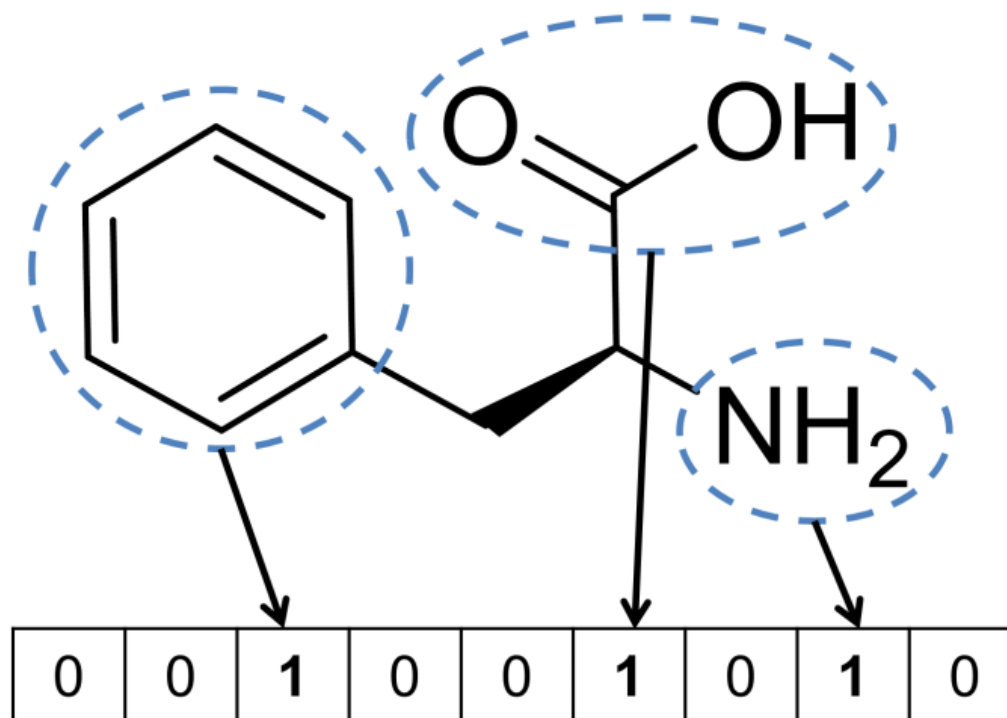
- Atoms: C, N, O, Cl, Br, ...
- Bonds:
 - Single –
 - Double =
 - Triple #
- Branches
 - Open (
 - Close)
- Ring numbers
 - Indicate ring open and close with integer



Chemical fingerprints (descriptor vectors)

1. Structural keys

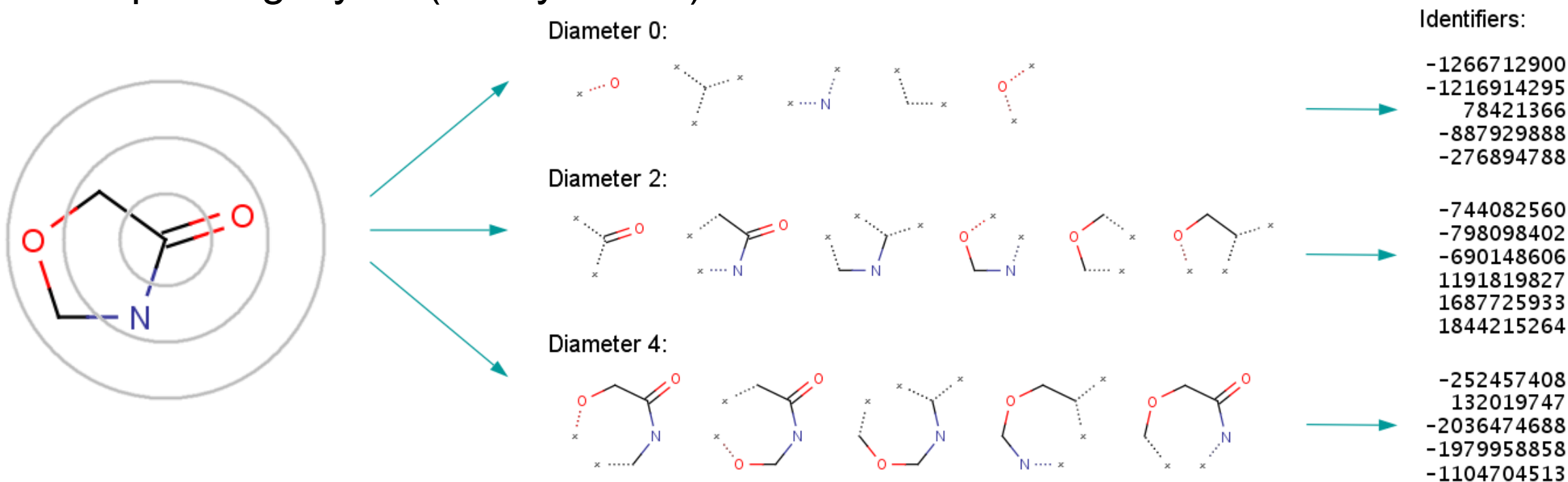
- Definition: **fixed bit strings** where each position = presence(1)/absence(0) of a predefined chemical group.
- Example: **MACCS** keys; common sizes **166** or **960** bits; used for similarity search.



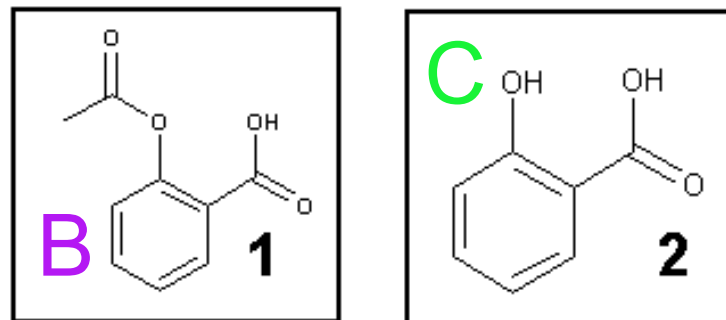
Chemical fingerprints (descriptor vectors)

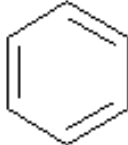
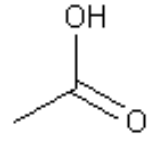
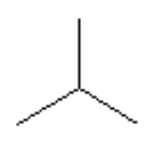
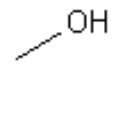
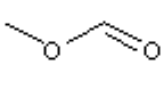
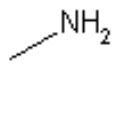
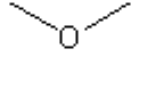
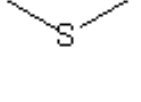
2. Hashed fingerprints

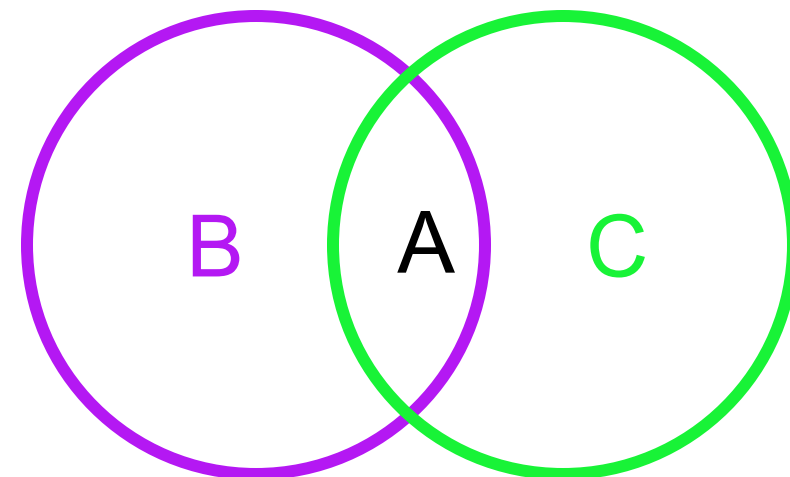
- Definition: vectors whose elements encode features **generated from the molecule** (not a fixed list); a **hash function** assigns patterns to bits. Length is user-set (e.g., 512/1024/2048).
- **Circular (ECFP/Morgan)**: encodes **atom neighborhoods** in expanding layers (heavy atoms).



Calculating Tanimoto similarity of fingerprints



1	1	1	0	1	1	0	1	0
2	1	1	0	1	0	0	0	0
								



A = Number of bits set in both = 3
 B = Number of bits set in (1), but not in (2) = 2
 C = Number of bits set in (2), but not in (1) = 0

$$\text{TANIMOTO COEFFICIENT} = A / (A + B + C)$$

$$= 3 / (3 + 2 + 0) = 0.6 \text{ or } 60\%$$

How you generate fingerprints has a large effect on the similarity value!

Please stop saying “The Tanimoto similarity is”

SIMILARITY

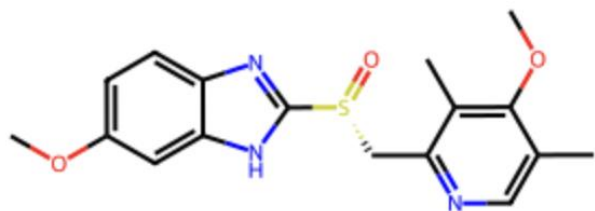
FINGERPRINTS

RANTS

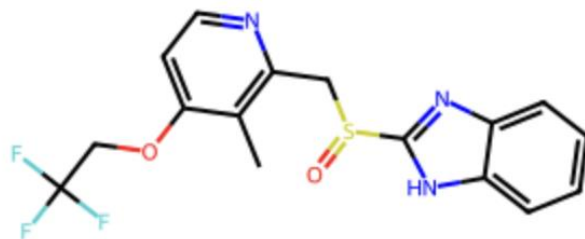
A simple tip to explain what you actually did

PUBLISHED

July 17, 2025



esomeprazole

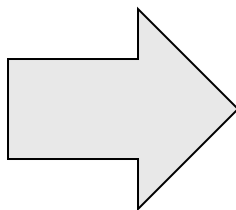
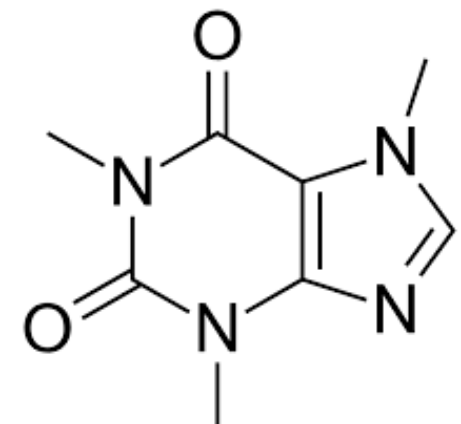


lansoprazole

RDKit fingerprint similarity: 0.79
Morgan fingerprint similarity: 0.43

2D Representations → Graphs

Adjacency matrix



	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
2	1	0	1	0	0	0	0	0	0	0	0	0	0	1
3	0	1	0	1	1	0	0	0	0	0	0	0	0	0
4	0	0	1	0	0	0	0	0	0	0	0	0	0	0
5	0	0	1	0	0	1	1	0	0	0	0	0	0	0
6	0	0	0	0	1	0	0	0	0	0	0	0	0	0
7	0	0	0	0	1	0	0	1	0	0	0	0	0	1
8	0	0	0	0	0	0	1	0	1	0	0	0	0	0
9	0	0	0	0	0	0	0	1	0	1	0	0	0	0
10	0	0	0	0	0	0	0	0	1	0	1	1	0	0
11	0	0	0	0	0	0	0	0	0	1	0	0	0	0
12	0	0	0	0	0	0	0	0	0	1	0	0	1	1
13	0	0	0	0	0	0	0	0	0	0	0	1	0	0
14	0	1	0	0	0	0	1	0	0	0	0	1	0	0

Index → Atom (role)

- 1 → CMe_1 (methyl carbon attached to N1)
- 2 → N1 (ring N; bonded to 1,3,14)
- 3 → C(=O)_A (carbonyl carbon; bonded to 2,4,5)
- 4 → O_A (carbonyl oxygen on 3)
- 5 → N3 (ring N; methylated)
- 6 → CMe_3 (methyl carbon attached to N3)
- 7 → C_ar_1 (aromatic C; in fused ring)
- 8 → N_ar_1 (aromatic N)
- 9 → C_ar_2 (aromatic C)

- 10 → N7 (ring N; methylated)
- 11 → CMe_7 (methyl carbon attached to N7)
- 12 → C(=O)_B (carbonyl carbon; bonded to 10,13,14)
- 13 → O_B (carbonyl oxygen on 12)
- 14 → C_ar_3 (aromatic C; closes rings to 2 and 7)

Property calculation

- Given a **2D graph** representation of a molecule you can calculate many properties.
- Calculations with **RDKit** - Research and development kit : The most popular open-source cheminformatic toolkit for molecular representation and manipulation.
- Properties/descriptors:
 - Molecular weight (MW)
 - **logP**
 - **Topological polar surface area (TPSA)**
 - H-bond donors/acceptors (HBD/HBA)
 - N rotatable bonds
 - **Structural alerts (PAINS/Brenk/ChEMBL alerts)**
 - Fingerprint similarity
 - **Quantitative Estimate of Drug-likeness (QED)**



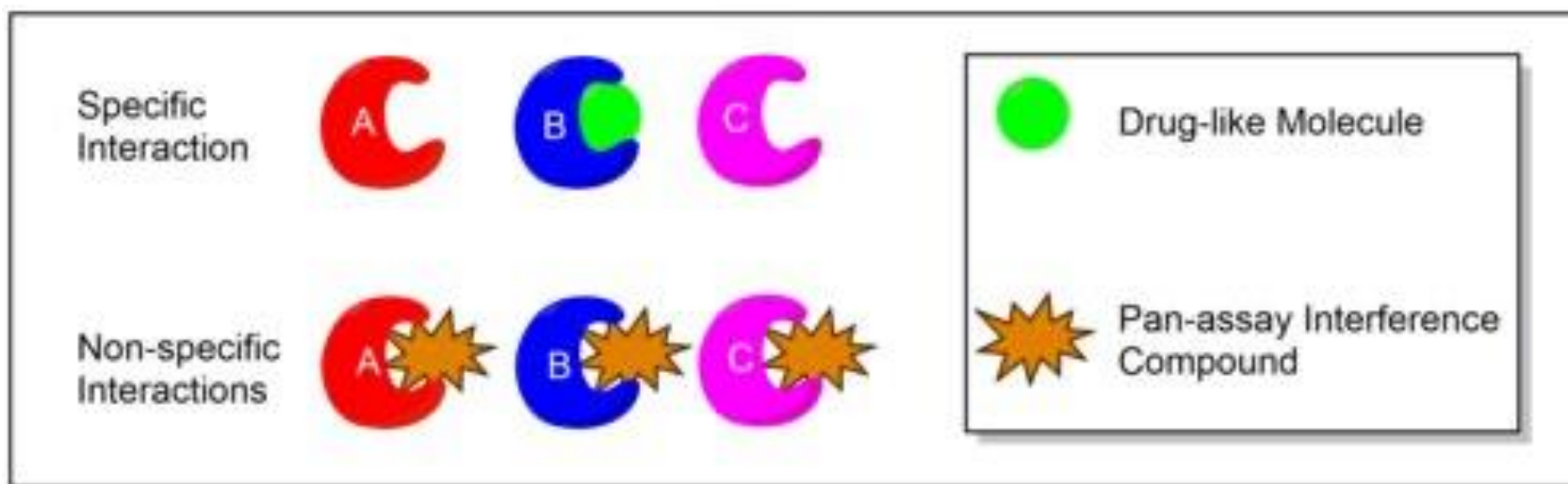
Open-Source Cheminformatics
and Machine Learning

A bit more detail on important descriptors for the practical

Descriptor	Explanation	Good range
logP	Octanol/water partition of the neutral form (lipophilicity & permeability proxy). In RDKit logP = Crippen cLogP, a fast 2D fragment-based estimate	sweet spot for oral drugs $\approx$ 1–3
Topological polar surface area (TPSA)	Sum of topological polar surface areas of heteroatoms (H-bonding/absorption proxy)	oral drugs usually <140 Å² CNS typically <90 Å²
Quantitative Estimate of Drug-likeness (QED)	Composite “drug-likeness” score from MW, logP, HBD/HBA, TPSA, rotatable bonds, aromatic rings & alerts → returns 0–1 score	≥0.6 is solid and ≥0.7 is strong

PAINS/BRENK alerts

- Flag substructures prone to **assay interference** or **nonspecific activity**
- **PAINS**: empirically frequent hitters from HTS; rich in *context-dependent* motifs.
- **BRENK**: broader **structural alerts** for reactivity/tox risk; more conservative, catches “chemically ugly” groups.

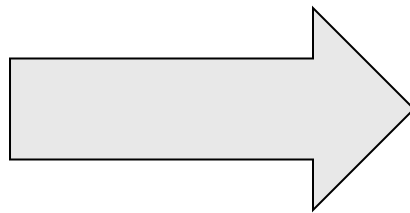
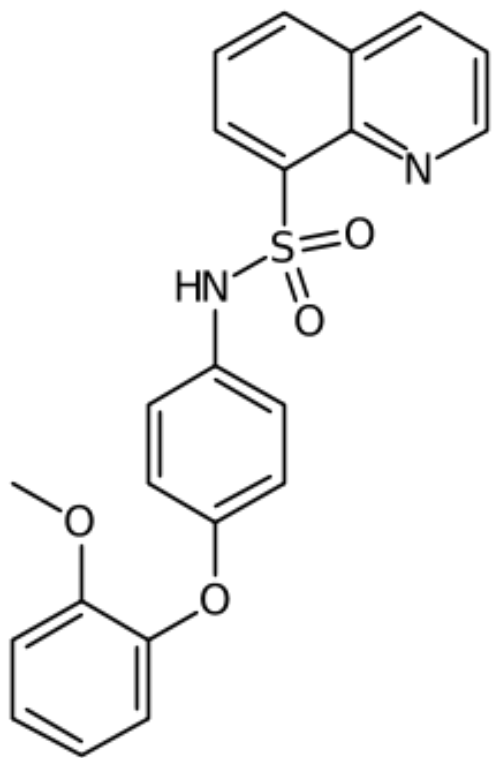


Any questions?

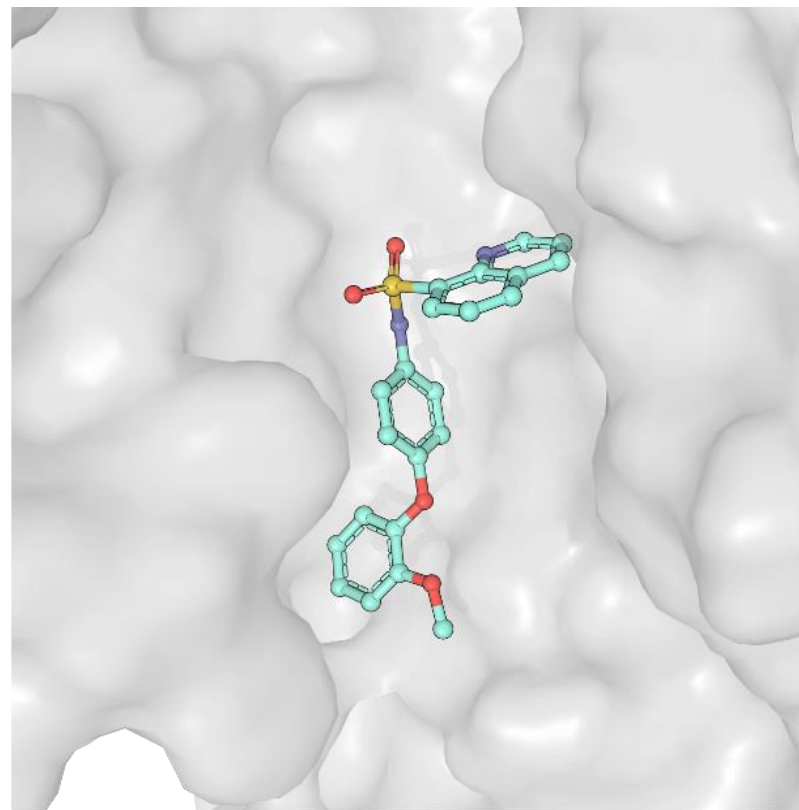
3D Representations

We exist in 3D, so do proteins + molecules

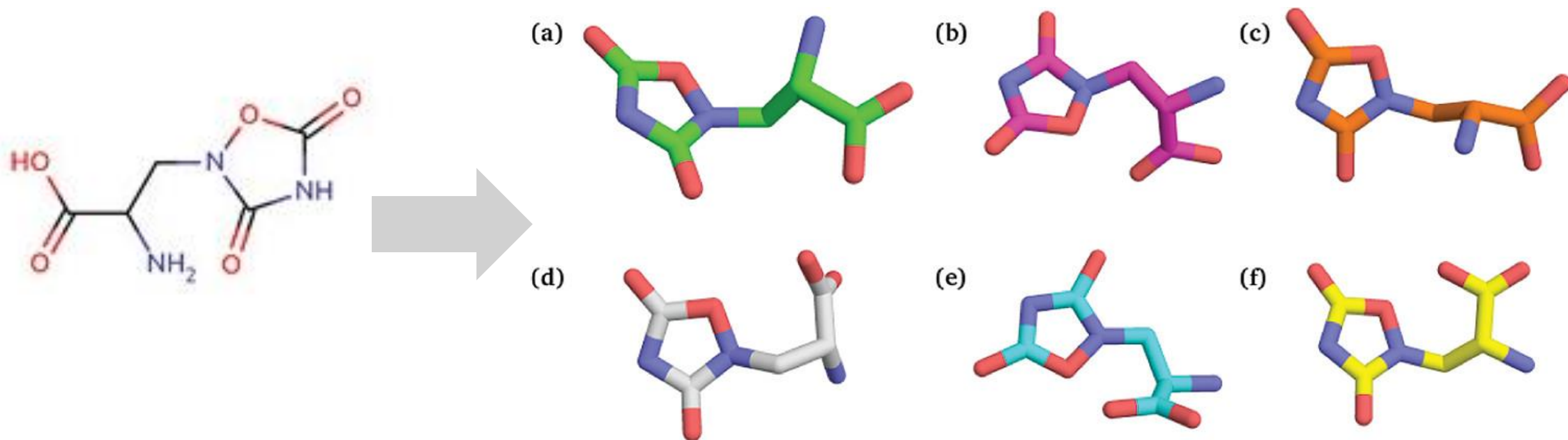
2D



3D



There can be many different 3D conformations of a molecule



- N possibilities grow ~exponentially with rotatable bonds (≈ 3 per bond), but the **relevant** low-energy set is usually tens (not thousands) for drug-like molecules
- Rings and special motifs can nudge that up or down

How to generate a 3D conformer ensemble → depends on what question you're trying to answer

- **Free conformers:** Given this molecule what are the possible conformations it can access *freely*?
- **Restricted conformers:** Given this molecule and a **similar molecule**, what is a predicted conformation?
- **Conformer in a binding site (docking):** Given this molecule and a protein, what is its **predicted site of binding and conformation**?

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Used in the practicals!

Calculate free (ligand-only) conformers with force fields

Procedure:

1. Build 3D from 2D → embed many conformers
 - RDKit uses by default Experimental-Torsion-Knowledge + Distance Geometry (**ETKDG**) which uses classical distance geometry

Calculate free (ligand-only) conformers with force fields

Procedure:

1. Build 3D from 2D → embed many conformers
 - RDKit uses by default Experimental-Torsion-Knowledge + Distance Geometry (**ETKDG**) which uses classical distance geometry
2. Assigning a force field, atom types & charges → minimize each conformer to a local minimum
 - Most popular force fields on RDKit are:
 - **MMFF94** (Merck): Small-molecule **organic** force field fit to high-level QM + empirical data; very good at **bonded terms and torsions** common in drug-like chemistry
 - **UFF** (Universal): Rule-based parameters that cover **most of the periodic table** (incl. metals/organometallics); designed for **coverage**, not peak accuracy

Riniker, S.; Landrum, G. A. “**Better Informed Distance Geometry (ETKDG)**” *J. Chem. Inf. Model.* **2015**.

Halgren, T. A. “**MMFF94: Basis, Form, Scope, Parameterization, and Performance**” *J. Comput. Chem.* **1996**.

Rappé, A. K.; Casewit, C. J.; Colwell, K. S.; Goddard, W. A.; Skiff, W. M. “**UFF: A Full Periodic Table Force Field**” *J. Am. Chem. Soc.* **1992**.

Application of free conformer generation: Posebusters (used in practical)



- PoseBusters: A package from OPIG which runs an RDKit-based test suite on predicted poses (docked compounds) to flag:
 - **Chemical/physical implausibilities** (stereochemistry/valence, bond lengths & angles, aromatic planarity, intra-ligand strain)
 - **Protein–ligand clashes**

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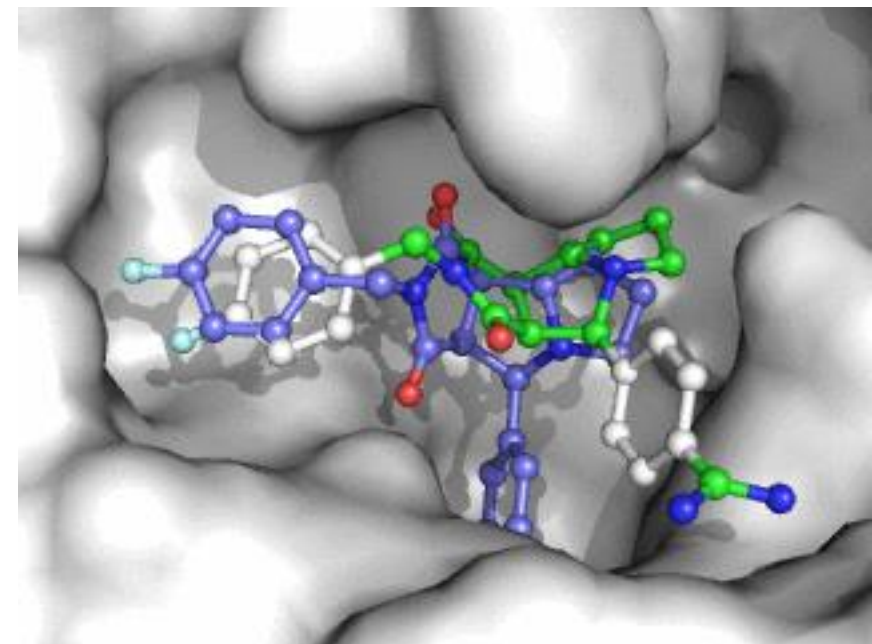
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 - **Protein–ligand clashes**

Energy ratio

The calculated energy of the input molecule is no more than 100 times the average energy of an ensemble of 50 conformations generated for the input molecule. The energy is calculated using the UFF³² in RDKit and the conformations are generated with ETKDGv3 followed by force field relaxation using the UFF with up to 200 iterations

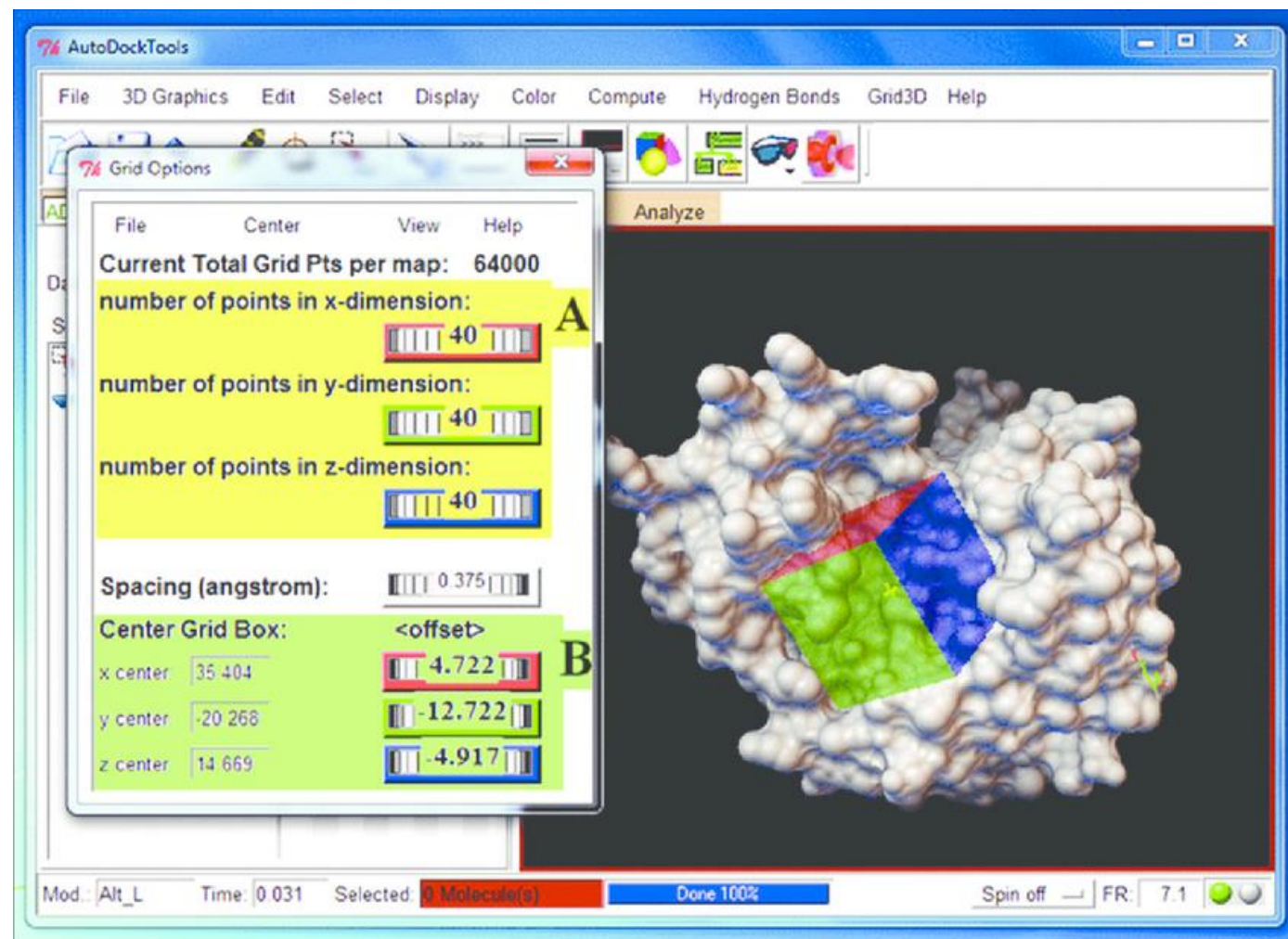
Calculate conformers in a binding site (docking)

- Protein-ligand docking uses a **sampling algorithm** and **scoring function**
- **Sample**: sample ligand poses in a binding site
- **Score**: approximate ΔG_{bind} with a fast function
- Typical assumptions: **flexible ligand**, **mostly rigid protein**, optional key side-chain/induced-fit tricks



The OG docking tool: Autodock Vina

- An open-source docking tool developed in the late 2000s
- **Smina** is fork of Autodock Vina which improved the scoring & minimization
- Smina also adds convenience functions which we use



Smina's sampling function

- **Search space:** 3D translation, orientation, and **ligand torsions**.
- **Global exploration (“exhaustiveness”):** runs multiple independent stochastic searches; step count per run scales with ligand flexibility; **promising intermediates are remembered**.
- **Local refinement:** after each perturbation, poses are **minimized** to the nearest local minimum of the scoring function.
- **Box-bounded sampling:** all moves occur **inside your docking box**.
- **Outputs = best minima:** keep an archive of low-energy solutions.

Default Vina-style scoring (used by Smina)

Score (negative is better!)

$$E = \sum \underbrace{e_{pair}(d)}_{\text{Protein-ligand atom-type pairs}}$$

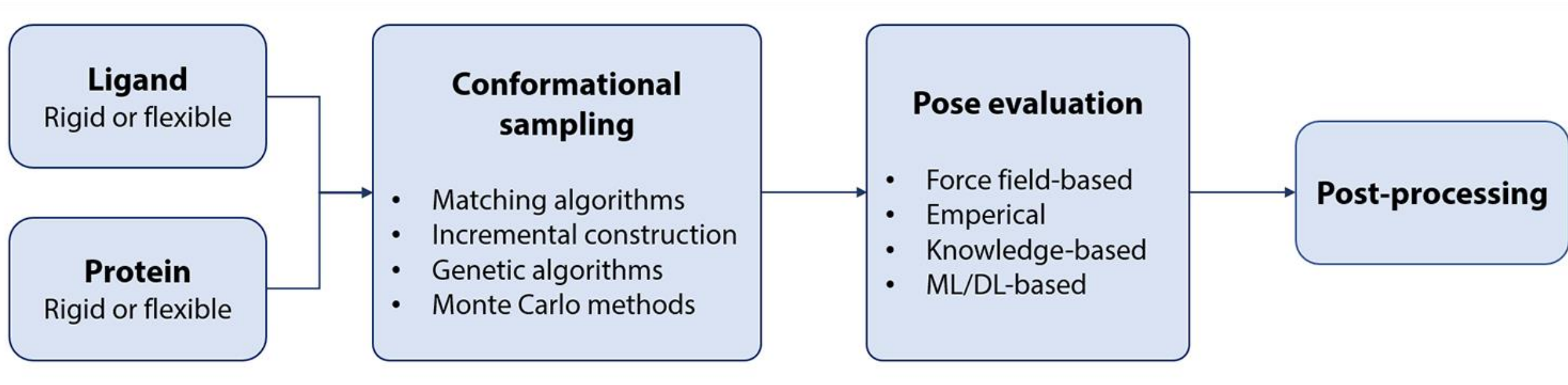
Surface
distance

$$d = r - R_i - R_j$$

Interatomic distance Atom i radius Atom j radius

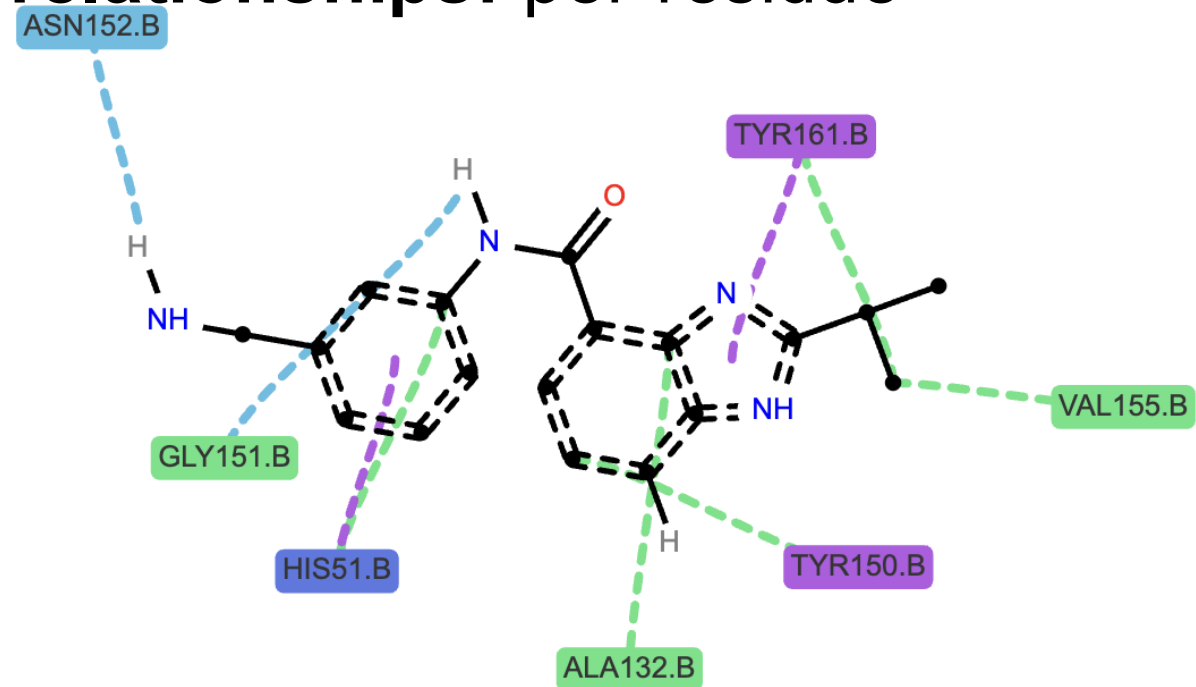
$$e_{pair}(d) = \left\{ \begin{array}{l} w_1 * Gauss_1(d) + \\ w_2 * Gauss_2(d) + \\ w_3 * Repulsion(d) + \\ w_4 * Hydrophobic(d) + \\ w_5 * HBond(d) + \\ w_6 * N_{\text{rotatable bonds}} \end{array} \right.$$

Typical docking workflow



Ok, but why go to all that trouble for 3D?

- **Rational drug design!!**
- **Fit & strain:** does a **low-energy** conformer fit? (minimize ΔE_{conf})
- **Explain how structure activity relationships:** per-residue contacts, hotspots, activity cliffs.
- **Selectivity:** exploit subtle pocket differences and avoid off-target effects.
- **Design vectors:** where to grow off a compound



Any questions?

Generative Methods

How do we design/generate the compounds we want to test?

2D

2D generative design: chemical language modeling

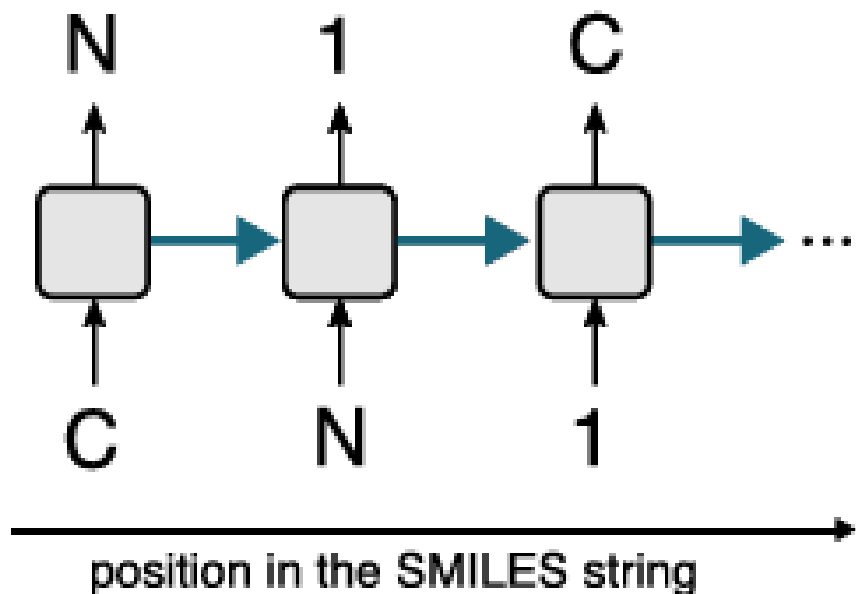
- Treat molecule design as a **sequence generation problem**

2D generative design: chemical language modeling

- Treat molecule design as a **sequence generation problem**

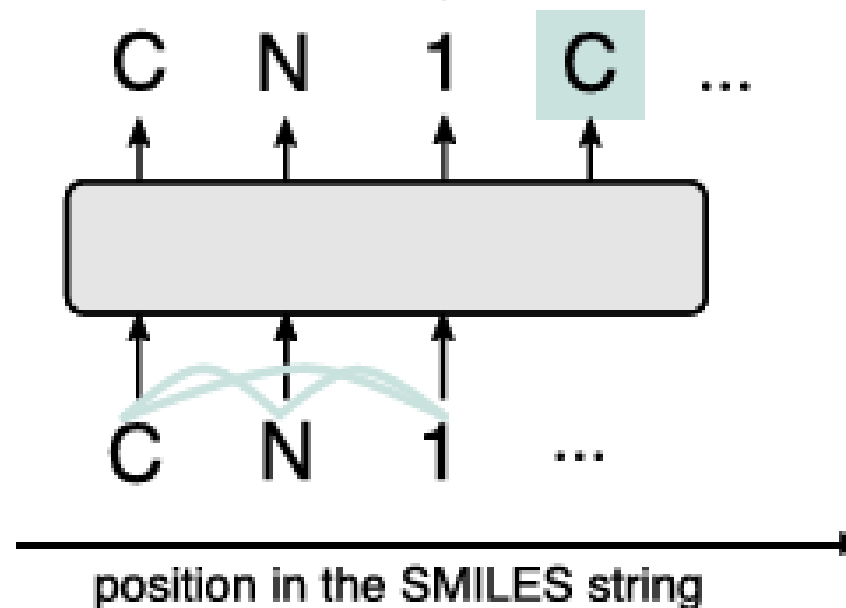
Recurrent Neural Network (RNN)

Use information from all previous tokens to predict the next



Transformer

Learn all pair relationships between sequence tokens to predict



2D generative design: chemical language modeling

- Treat molecule design as a **sequence generation problem**

Recurrent Neural Network (RNN)

Use information from all previous tokens to predict the next

- + Light, simple
- + Strong local syntax/validity
- + Modest data needs
- Non-parallel, slower
- Poor long-range memory
- Vanishing gradients/mode collapse

Transformer

Learn all pair relationships between sequence tokens to predict

- + Global context, long-range
- + Parallel, scales well
- + High diversity/novelty
- Data + compute hungry
- Memory heavy for long SMILES
- Needs positional encoding, validity tuning

2D generative design: chemical language modeling

- Treat molecule design as a **sequence generation problem**

Recurrent Neural Network (RNN)

Use information from all previous tokens to predict the next

Transformer

Learn all pair relationships between sequence tokens to predict

Seminal papers

Segler MHS; Kogej T; Tyrchan C; Waller MP.
ACS Cent. Sci. (2018).

doi:10.1021/acscentsci.7b00512.

Olivecrona M; Blaschke T; Engkvist O; Chen H.
J. Cheminf. (2017). doi:10.1186/s13321-017-0235-x.

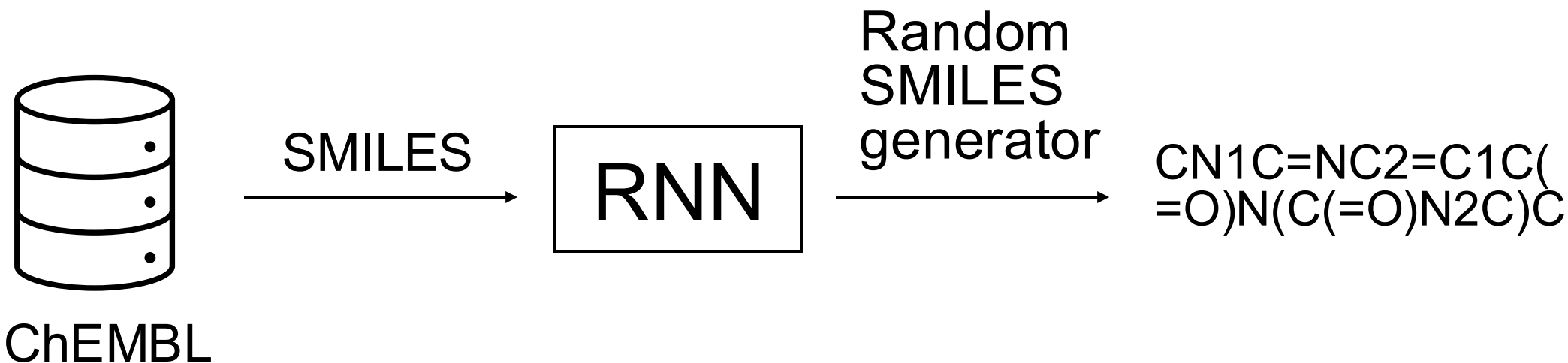
Zhumagambetov R; Molnár F; Peshkov VA; Fazli S. **RSC Adv.** (2021).

doi:10.1039/D1RA03086H.

Bagal V; Aggarwal R; Vinod PK; Priyakumar UD. **J. Chem. Inf. Model.** (2021).
doi:10.1021/acs.jcim.1c00600.

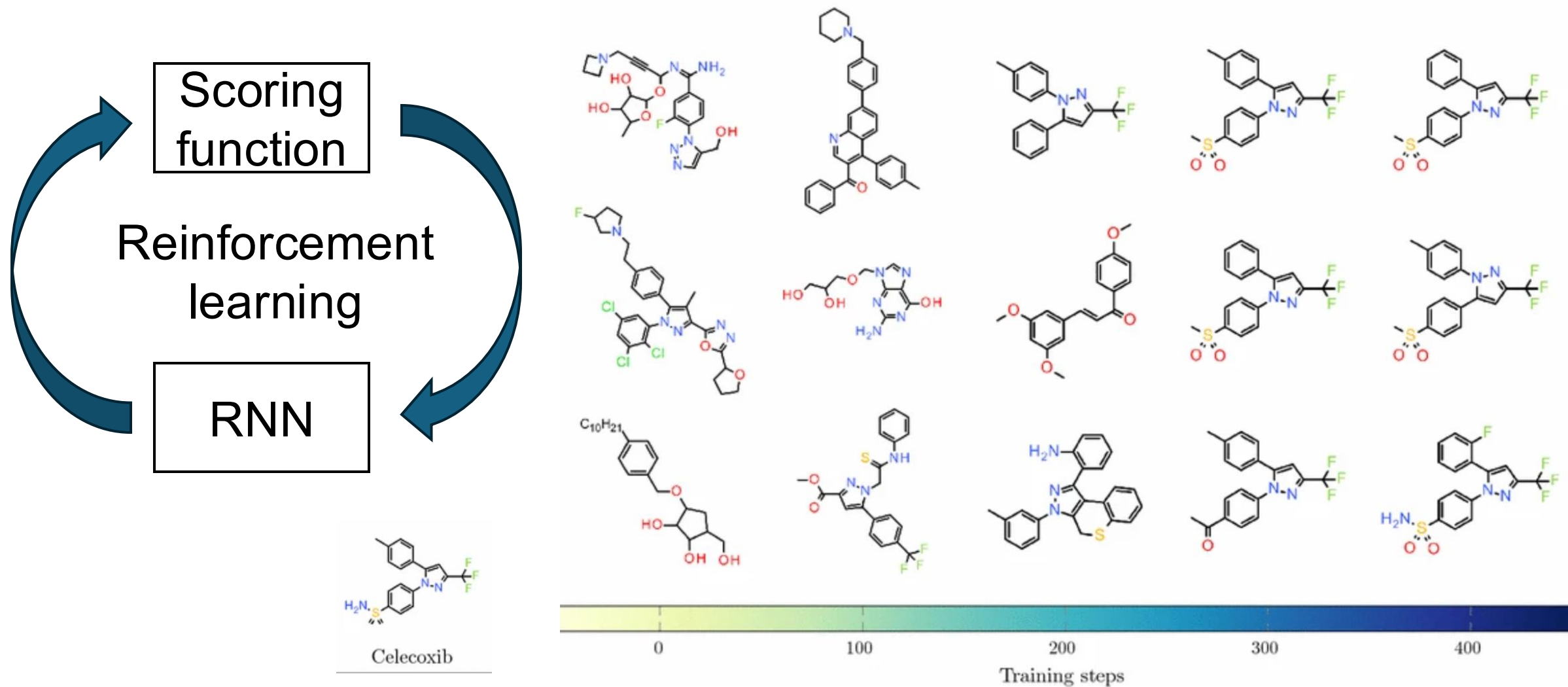
REINVENT: A RNN steered with reinforcement learning

- **Prior:** SMILES RNN trained on ChEMBL (use **randomized SMILES** for augmentation).
- Learns SMILES syntax + drug-like distribution → **valid, diverse** molecules.
- Serves as **unbiased random SMILES generator** (sample via temperature/top-k).
- This prior is later **steered with reinforcement learning** (Agent + scoring)

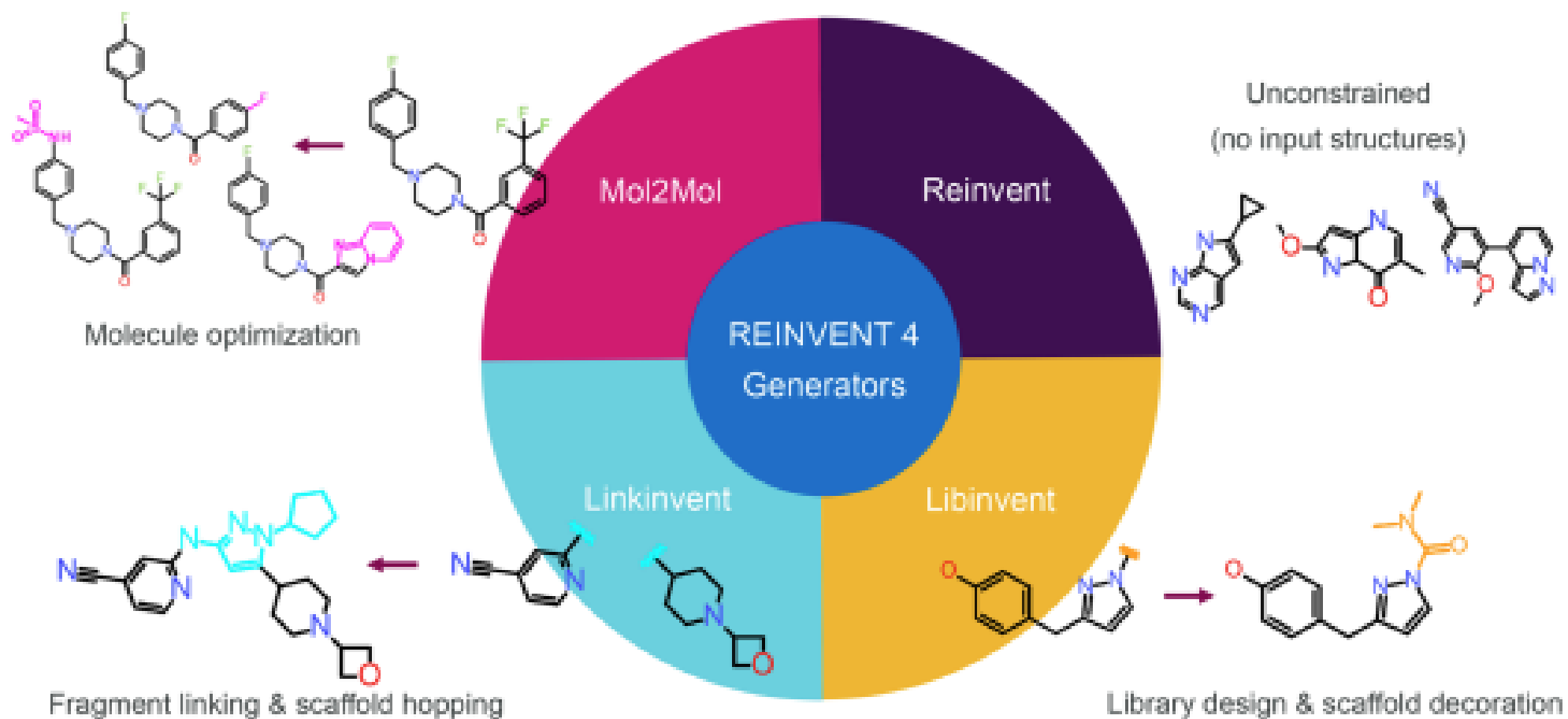


(3 million molecules)

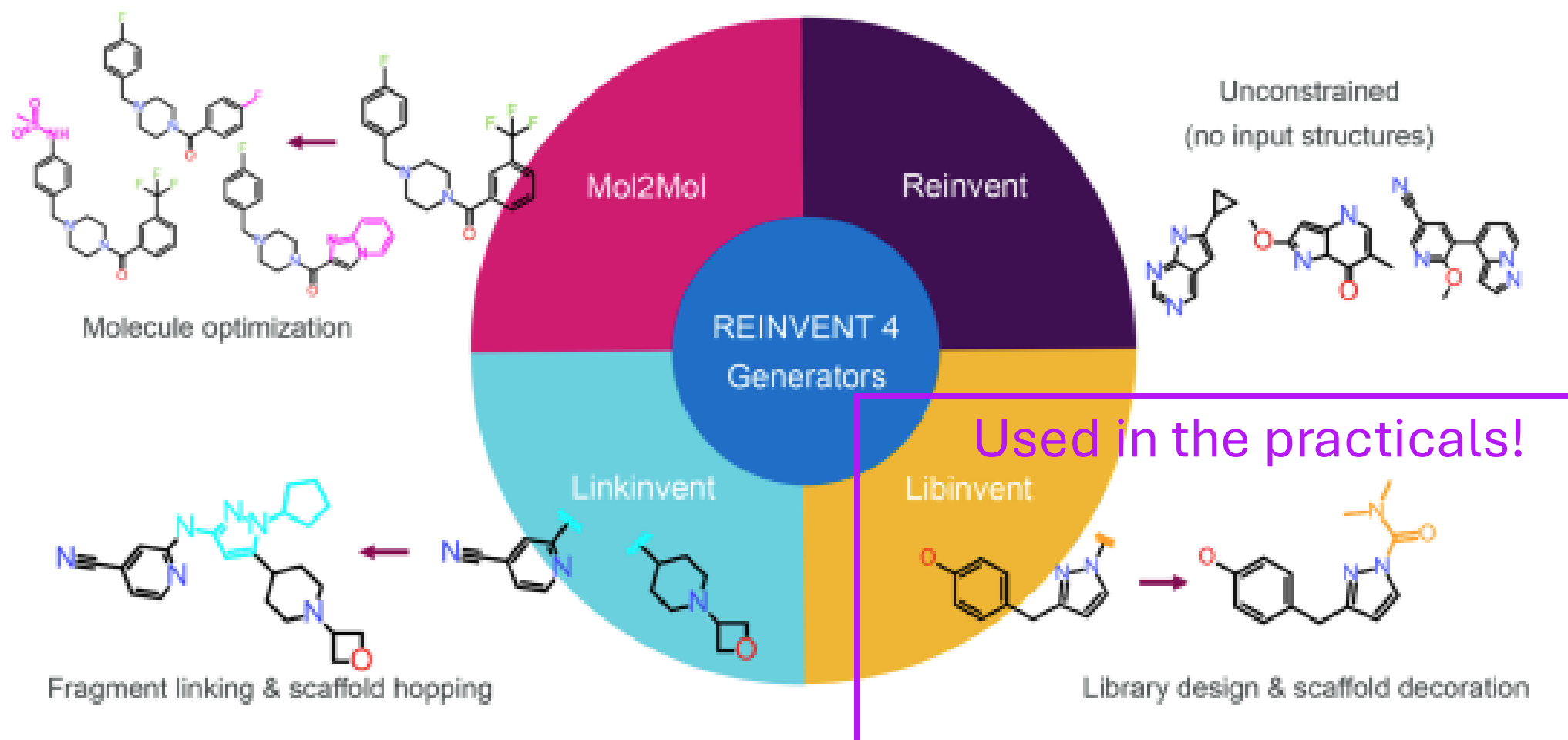
Reinforcement learning guides generation to desirable space



Flavors of REINVENT



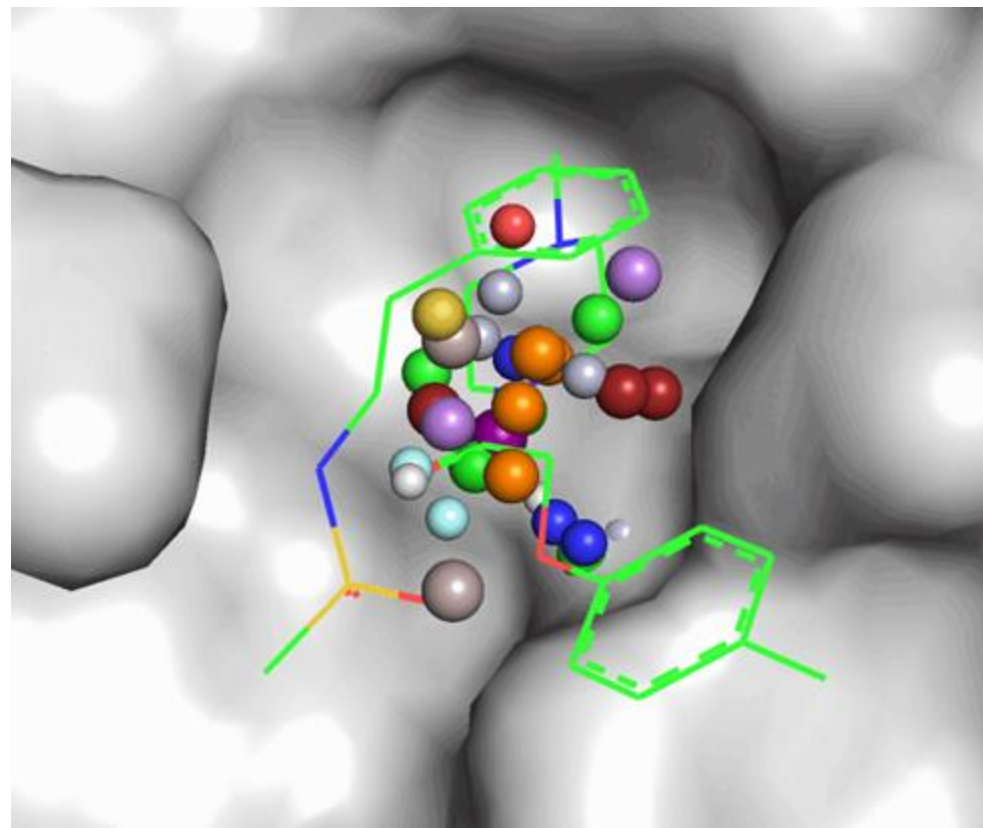
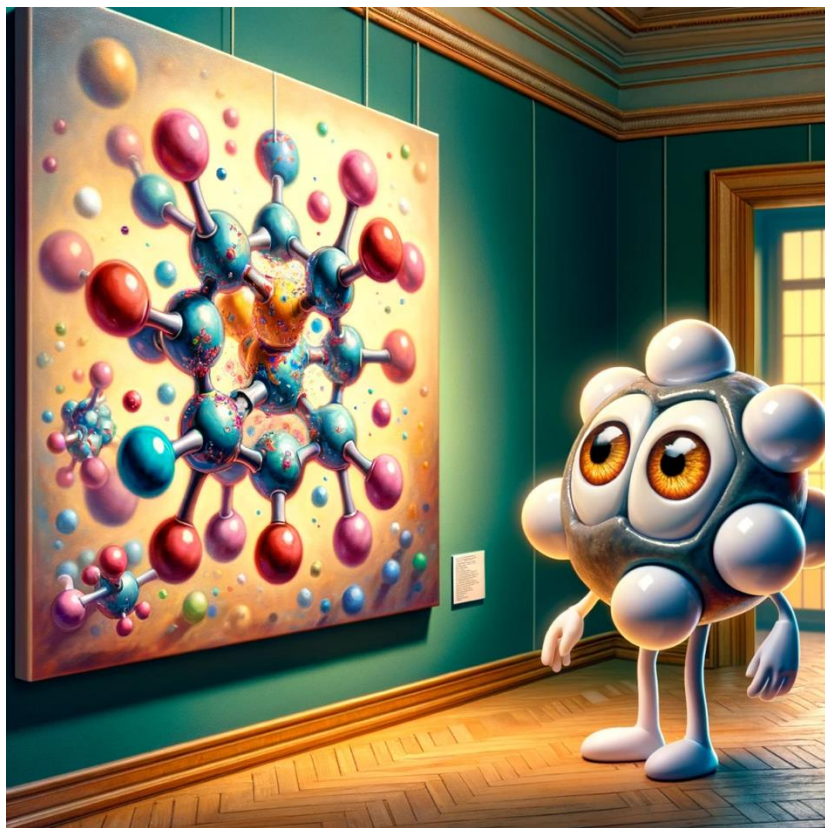
Flavors of REINVENT



3D

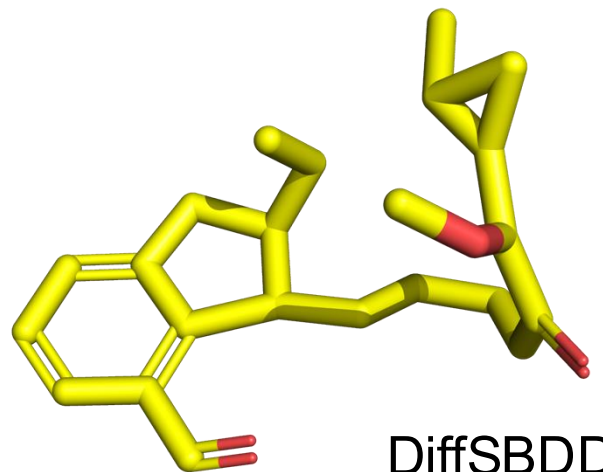
Can compounds be generated in the pocket?

3D diffusion models can generate molecules

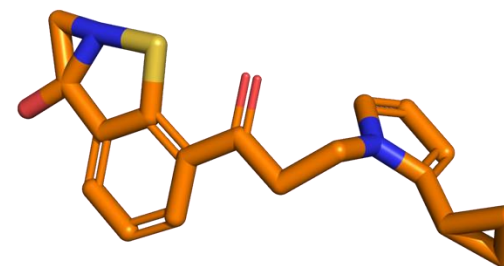


Current diffusion models generate chemically invalid molecules

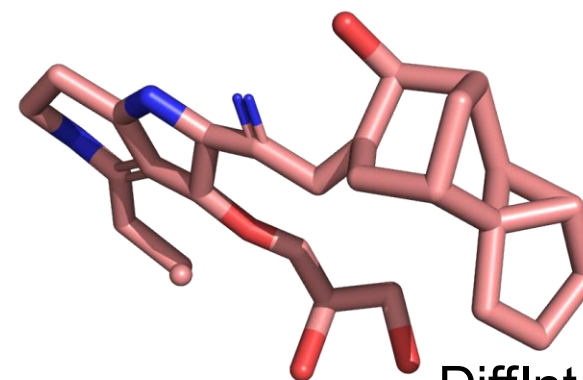
- **High internal steric clash**
- **Unrealistic geometries**
(strained bonds, torsions)
- Miss key protein–ligand **interactions**
- **Poor drug-likeness (QED, PB-valid)**



DiffSBDD



ShEPHERD



DiffInt

“Deep learning methods fail to generate structurally valid molecules and 3D conformations”

Benchmarking 3D Structure-Based Molecule Generators

Natasha Sanjrani,* Damien E. Coupry, Peter Pogány, David S. Palmer, and Stephen D. Pickett

Cite This: *J. Chem. Inf. Model.* 2025, 65, 8006–8021

Read Online

- **Task 1: Blind set:** 20 diverse protein targets, pass if key interactions in $\geq 50\%$ molecules; also checks similarity to known inhibitors and effects of metals/cofactors
- **Task 2: Target-specific checks:** Bespoke interaction tests on selective and pan-active targets; pass requires $\geq 50\%$ interaction recreation
- **Task 3: Virtual screening:** Pass if $\geq 10\%$ of molecules recreate interactions and average Tanimoto > 0.5 to known hits (RDKFingerprint).

Overall pass rates are low

Table 2. Fraction of Passed Tests per Task for Each Evaluated Method after Compound Filtering through MOSES, PoseBusters and Additional Allene and Fused Ring Checks^a

	task 1	task 2	task 3
Pocket2Mol	3/20 (1069)	2/4 (857)	2/2 (8282)
PocketFlow	1/20 (32,500)	1/4 (37,500)	0/2 (240,000)
DiffSBDD	3/20 (21,261)	1/4 (32,306)	2/2 (215,106)
MolSnapper	3/20 (12,541)	1/4 (13,719)	2/2 (96,655)
AutoGrow4	3/20 (813)	0/4 (192)	1/2 (719)
LigBuilderV3	0/20 (5425)	0/4 (650,990)	0/2 (3,832,963)

^aThe required percentage of recreated interactions in 50% for tasks 1 and 2 and 10% for task 3. The total number of compounds generated post-filtering for each task are shown in brackets.

What about co-folding?

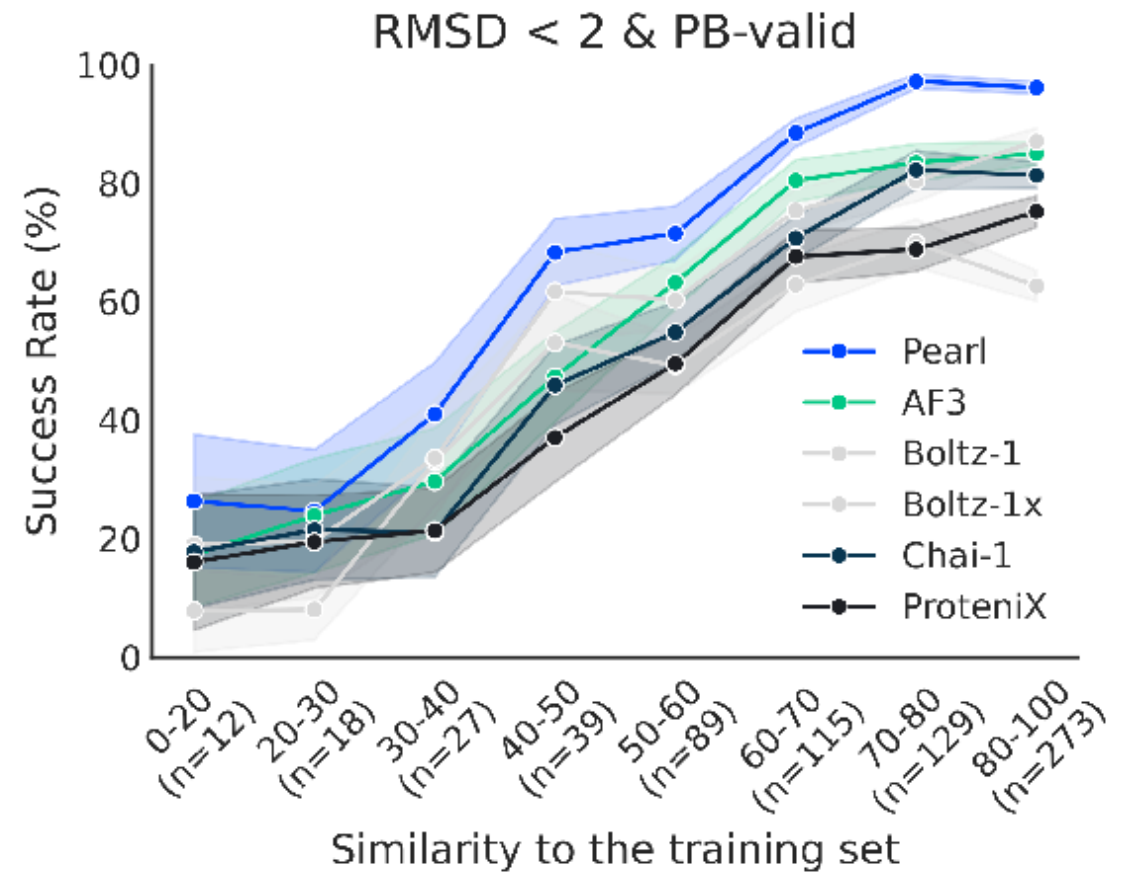
- Predict a protein structure AND bound ligands

We Still Haven't Found What We're Looking For - The Continuing Evolution of Protein-Ligand Co-Folding Methods

6 minute read

 **Published:** November 03, 2025

<https://patwalters.github.io/Cofolding-Evolution/>



Genesis Research Team *et al.* *Genesis Molecular AI – Technical Report*, Oct 28, 2025.

Škrinjar P; Eberhardt J; Tauriello G; Schwede T; Durairaj J. **bioRxiv** (2025). doi:10.1101/2025.02.03.636309.

The state of 3D generation

- The field is heading in this direction and **showing initial promise**
- But we still see a **lack of conformational validity and interaction recovery**
- **Synthetic accessibility** is still a **huge issue** (if the compound can't be synthesized or synthesized, it will never be tested)
- Co-folding methods **struggle with generalization**

Practicals

Putting everything together to cure Zika!

Putting it all together!

Morning

- **Practical 1:** Introduction to RDKit
- **Practical 2:** Introduction to REINVENT

Afternoon

- **Practical 3:** An example workflow for designing a Zika NS2B-NS3 protease inhibitor using LIBINVENT
- **Practical 4:** Choose your own adventure to design **the best 10 compounds** to cure Zika

Zika

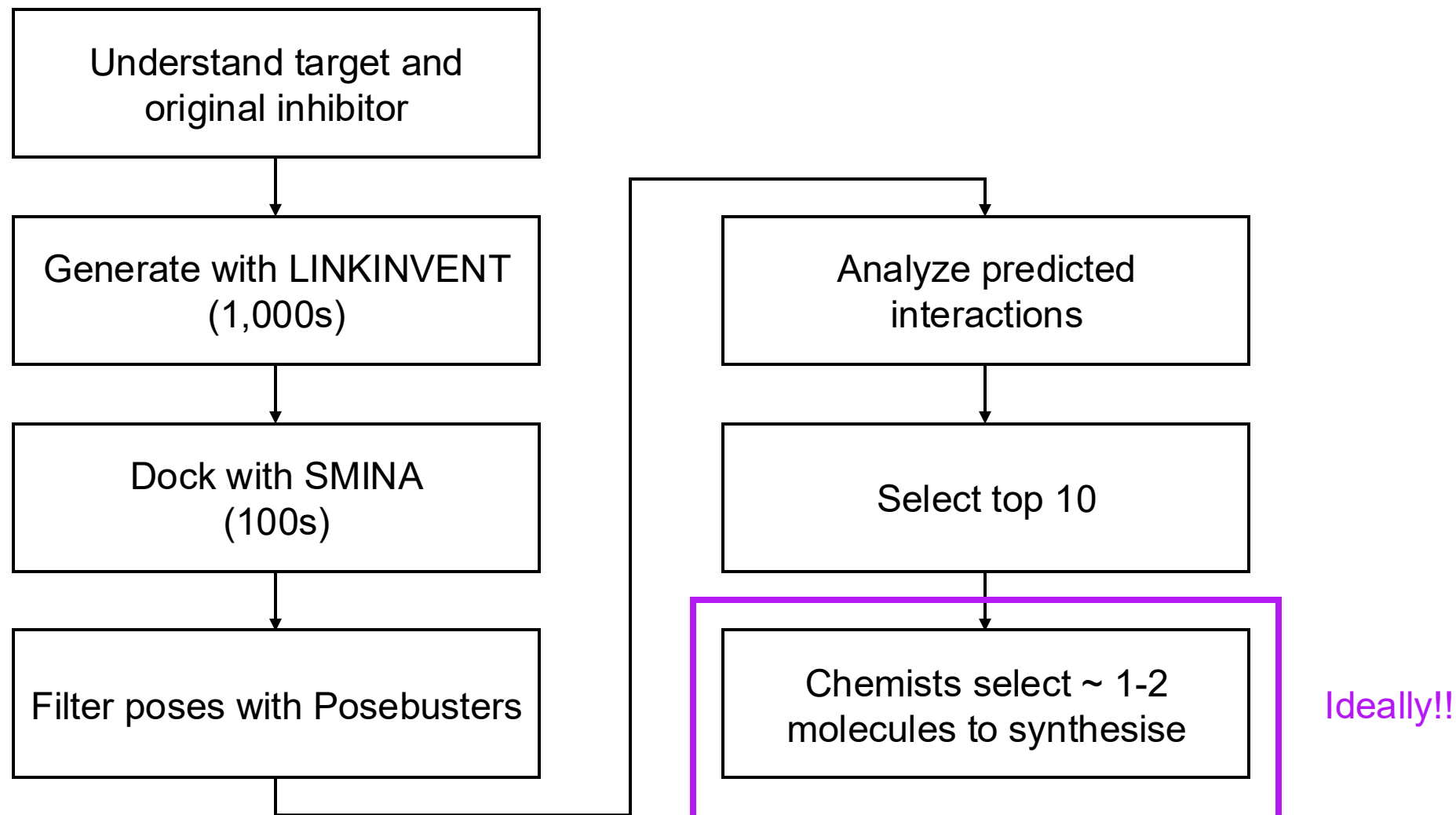
- Virus transmitted through mosquitoes
- Infection is associated with:
 - **Microcephaly in newborns** from infected mothers
 - Neurological diseases in adults - **Guillain-Barre syndrome**
- **No drugs** or vaccines currently approved for use



Our target: Zika NS2B-NS3 protease

- NS2B-NS3 protease is **essential in the viral replication process**, therefore **developing an inhibitor** that binds in the active site could **disrupt replication**.
- We have a crystal structure of an inhibitor bound to the protease (PDB: **7I9O**) with micromolar affinity published in **October 2025**.
- You'll try to use part of the original inhibitor to **design a better one!**
 - Better being:
 - Better drug-like properties
 - More hydrogen bond interactions
 - Better docking score

Practicals 3 + 4: Design workflow



Backup slides

Diffusion

